SEVARO AT RUTGERS HEALTH HACK

Challenge 1 Handbook

What the problem is, the issues to plan for, how your tool will be tested, and what a strong answer does.

THE SHORT VERSION

Sevaro's neurologists look after patients at more than 200 hospitals over video, and every hospital does things its own way: which clot-busting drug it uses, where the video exam happens, who to call for a blocked artery, what to title the note, whether a charge has to be entered. A real hospital guide has more than 80 of these rules.

How it's handled today. In the middle of a stroke case, the doctor has to stop, open the hospital's guide, find the right section, and come back. Under time pressure they often go from memory instead, and memory is often wrong for that hospital. The mistakes usually only get caught after the case is closed.

Teams build a companion that **puts the right rule for this hospital in front of the doctor at the moment they need it**, inside the screen they're already working in, without being asked.

CHAPTER 1

The situation

Sevaro's doctors work in **Synapse**, Sevaro's platform for calls, video and notes. While they treat a patient they fill in the **caselog**: a form that walks them through the case step by step (intake, history, imaging, recommendation, note, sign-off).

Each hospital has a **guide** listing its own rules, field by field. Some hospitals also have a short **storyboard**: a summary card shown at the top of the caselog. It's kept separately, so it can disagree with the guide.

Picture a stroke case. The doctor is at the imaging step and marks that the patient has a blocked large artery. Right now they need to know: can this hospital remove the clot, or does the patient need to be moved somewhere that can? And who makes that call? The answer is different at every hospital, and at some it depends on the time of day.

Synapse, in brief

Synapse is the platform Sevaro's doctors work in all day. At a high level, it handles:

- **Phone and video consults** with the hospital and the patient.
- **The caselog:** the step-by-step form for each case, from intake to sign-off, with the hospital's storyboard card shown at the top.
- **Documentation and charting:** the doctor's note for each case.
- **Clinical decision support:** tools that help the doctor during the case.
- **Rounding lists:** scheduled follow-up visits with patients already in the hospital.

Where your tool fits. Think of your tool as a companion that sits inside the caselog. Each time the doctor moves to a new step or changes a field, Synapse hands it the case details (Chapter 5 lists them). Your tool looks up that hospital's guide and decides whether to show something for that field. It only reads guides; it never changes them, and it never stops the caselog working. You won't have access to the real Synapse, so you'll build a stand-in (see Chapter 3).

Who's involved

THE DOCTOR

Teleneurologist

Treating a patient remotely, filling in the caselog, often in a different time zone from the hospital.

THE RULES

Hospital guide

Each hospital's settings, one field at a time, each with the date it was last checked.

A SECOND SOURCE

Storyboard

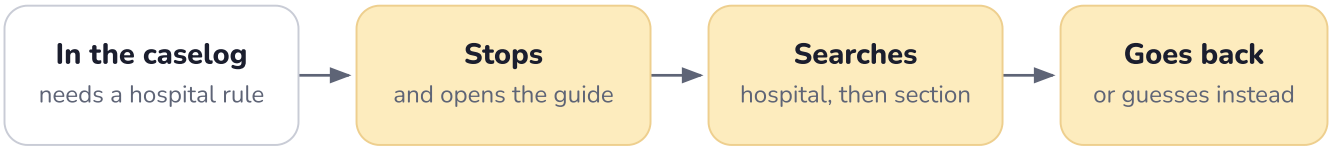
A short summary card in the caselog. Usually agrees with the guide. Sometimes doesn't.

AFTERWARDS

Quality team

Finds and follows up the mistakes once a case is closed. The goal is to stop them happening at all.

How it's handled today



The amber steps are the ones your tool should remove.

Don't build just a chat bot. The main thing is nudges that appear on their own, for the right field, at the right step. Answering typed questions is a second feature. A chat assistant on its own will not place highly.

CHAPTER 2

The issues to plan for

Getting the right rule to the doctor at the moment they need it depends on many things at once. **Before you build, list them yourselves**, then check your list against these. Each comes with an example and what to anticipate, because the hospital you're tested on won't be one you practiced on.

1. Every hospital is different

The same question has a different answer at each hospital.

Example: One hospital uses Alteplase, another uses Tenecteplase, and they have different order sets to search for.

Anticipate: Answer only from the guide of the hospital in this case. Never fill a gap with another hospital's answer.

2. When a rule is needed

A rule is only useful at the step where it matters. The caselog tells you which step the doctor is on and what they just changed.

Example: The doctor marks "large vessel occlusion present = Yes" at the imaging step. That's when the hospital's LVO rule should appear.

Anticipate: Drive nudges from the case details you're given, not from a fixed script. Keep the link between caselog fields and guide fields as data you can change.

3. Knowing when to stay quiet

With 80 or more rules per hospital, showing everything trains doctors to ignore all of it.

> **Example:** The doctor enters "no allergies". Nothing in the guide applies, so nothing should pop up.

Anticipate: Be ready to show that silence is deliberate, and to explain why each nudge that interrupts deserves to.

4. The hospital's local time

Many rules depend on the day and time at the hospital, and the doctor is often somewhere else.

> **Example:** A hospital can remove clots on weekdays until 5pm its time. It's 5:30pm there and 7:30pm where the doctor is. The answer is "outside the window", so the patient goes elsewhere.

Anticipate: Always work in the hospital's own time zone, using a proper time zone library rather than fixed hour offsets.

5. Times and days written in words

Hours and days in the guides are written as text, in many different styles.

> **Example:** "Mon-Sat 0700-1900", "Tue/Thu 0800-1500", "Daily", "24/7", "weekend on-call", "As scheduled".

Anticipate: Expect styles you haven't seen yet. If your tool can't be sure what a phrase means, it should say so rather than guess.

6. Information that isn't there

Some hospitals' guides are missing fields that others have.

Example: A doctor asks which note title is required, and this hospital's guide doesn't list one.

Anticipate: Say "Not available in the hospital guide." Never guess, and never borrow the answer from a hospital that does have it.

7. Information that may be out of date

Every field records when it was last checked. Anything checked more than 180 days before the case may no longer be true.

Example: A field was last checked seven months before the case. Show its value, clearly marked as possibly out of date.

Anticipate: Work it out from the dates in the data every time. Any field, at any hospital, can be stale.

8. Two sources that disagree

The guide and the storyboard are kept separately, so they can say different things.

Example: The guide says do the stroke video exam in the CT holding bay. The storyboard says the patient's room.

Anticipate: Check every field that has a storyboard entry. When they disagree, show both with their sources and don't pick one. When they agree, don't raise a false alarm.

9. Rules that leave something out

Some rules depend on hours the guide never gives.

Example: "ED pharmacist prepares during staffed hours; nursing mixes overnight", with no staffed hours listed.

Anticipate: Say what is unknown and give both possibilities. Don't invent a cut-off time.

10. When to block

A hard stop blocks the doctor until they fix something or acknowledge it. Used too often, it gets in the way; used too rarely, mistakes slip through.

Example: A hospital requires a charge before sign-off and none has been entered, so Complete is blocked until it is.

Anticipate: Only block when the guide truly requires it, and be ready to explain why.

11. One rule can switch another off

Some rules only matter if another rule allows them.

Example: At a hospital where the remote doctor doesn't enter medication orders, showing them the order-set name to search for would be wrong. The useful nudge is that the hospital's own doctor places the orders.

Anticipate: Check how fields relate to each other before deciding what to show, and watch for values that mean "no" or "not applicable" in words.

12. New hospitals and new fields

Hospitals get added all the time, and guides gain fields.

Example: A new hospital's guide arrives with a field none of the others have.

Anticipate: Adding a hospital or a field should be a data change, never a code change. Expect field names you haven't seen.

13. When the guide can't be reached

Guidance helps the doctor, but it must never stop them working.

Example: The guide data fails to load at the start of a case.

Anticipate: Keep the caselog working, and tell the doctor plainly that guidance isn't available.

CHAPTER 3

What to build

- **Nudges that appear on their own (the main thing).** Your tool watches what the doctor is doing in the caselog and shows the relevant hospital rule for the right field, without being asked.
- **Answers to typed questions (second).** The doctor can also ask in plain words, like "Who mixes the drug here?", and get an answer taken from that hospital's guide.

You'll need a simple stand-in caselog to show the nudges in. It doesn't have to look like Synapse.

Three kinds of nudge

KIND	WHAT IT DOES	EXAMPLE
Hard stop	Blocks the action until the doctor fixes it or acknowledges it.	A required charge must be entered before sign-off.
Advisory	Shown to the doctor, but doesn't block.	This hospital can't remove clots; the ED doctor starts the transfer.
Info	Available if the doctor looks for it; doesn't interrupt.	CT perfusion isn't available here.

Every nudge and answer must show

- **Which hospital and which guide field** it came from.
- **Its status**: answered, not in guide, time-dependent, stale (possibly out of date), conflict (sources disagree), or hard stop. Chapter 5 explains each.

One case, start to finish

This is the example case from the kit (encounter_context_example.json, sample case N2), walked through step by step.

1. **What your tool is told:** Highland Valley Hospital (SYN-006), a stroke case, imaging step, the doctor has just set "large vessel occlusion present" to Yes. It's Tuesday 7:30pm at the hospital (Chicago time); the doctor is in New York, where it's 8:30pm.
2. **Which rule applies:** the caselog map links that field to the hospital's LVO Protocol, to be checked against local time.
3. **What the guide says:** "Thrombectomy capable Mon-Fri 0700-1900. During coverage, teleneurologist activates the synthetic EVT team. Outside coverage, ED provider initiates transfer to a regional comprehensive stroke center."
4. **Working it out:** 7:30pm on a Tuesday, hospital time, is outside the 7am to 7pm window. (Using the doctor's 8:30pm would give the same answer here, but only by luck.)
5. **What appears:** an advisory: outside the thrombectomy window, so the ED doctor starts a transfer. It must not suggest activating the EVT team. It shows its source (SYN-006, LVO Protocol) and its status (time-dependent).

What a strong one looks like

The doctor never leaves the caselog. At each step, the one rule that matters for this hospital, at this time, appears when it's needed, with where it came from. It's quick to take in during a stroke case, and everything else stays out of the way.

CHAPTER 4

The test: a new hospital

During your demo we will give you two files that look just like `hospital_guides.json` and `storyboard_notes.json`, for a hospital you haven't seen. **This is what we test you with:** you drop them into your tool. Your system needs to be ready to take in data of these types, pick up the new hospital, and then show the right nudges and answer questions about it, each with its source and status.

How it compares with the kit. The files are comparable to what you already have: the same layouts and the same kinds of fields. What's different is the hospital itself: a new name, time zone and set of rules, with its own gaps, dates and quirks, just as each of the seven sample hospitals has its own.

So the heart of this is a rules engine, not a list of answers. It should read whatever guide it's given, work out what applies to the case in front of it, and decide whether to speak up. The screen just shows what the engine found.

- Build for hospitals you haven't seen. A tool tuned to the seven sample hospitals will break on an eighth.
- Real guides are messy in all the ways described in Chapter 2. Your engine should cope with every one of them, every time.

For each nudge or answer, your tool should say

QUESTION	FOR EXAMPLE
What's the answer?	The rule, in plain words.
Where did it come from?	The hospital and the guide field (or the storyboard).
How sure is it?	The status: answered, not in guide, time-dependent, stale, conflict or hard stop.
What should the doctor do?	Including whether they're blocked until they act.

What the brief asks you to show

1. Nudges driven by the case details, covering the sample scenarios N1 to N9, including staying quiet in N8.
2. Answers that pass the sample questions V01 to V20, each with its source field and status.
3. Rules worked out in the hospital's local time, including when the doctor is in a different time zone.
4. Out-of-date, conflicting and missing information shown openly, never hidden.
5. A new hospital guide in the same format, added during your demo, with questions and scenarios you haven't seen.

Easy ways to lose ground

- Building only a chat box.
- Giving an answer from the wrong hospital.
- Guessing when the guide doesn't say.
- Using the doctor's clock instead of the hospital's.
- Quietly picking one side when the guide and storyboard disagree.
- Popping up on every field, or blocking when you don't need to.
- Needing a code change to add a hospital.

CHAPTER 5

The rules, in short

All hospitals, rules and names here are made up for the hackathon. This is not clinical guidance.

What your tool is told about each case

Every time the doctor moves to a new step or changes a field, your tool gets the case details. Use all of them, not just the hospital.

DETAIL	EXAMPLE	WHY IT MATTERS
Hospital	SYN-006	Which guide applies.
Consult and encounter type	Stroke, acute	Some rules only apply to stroke cases, or to rounding visits.
Caselog step	Imaging	Where the doctor is in the case.
Field change	Ivo_present = true	What just happened that might need a nudge.
Hospital's local time and time zone	19:30, Chicago	Many rules depend on day and time at the hospital.
Doctor's time zone	New York	The doctor is often in a different zone.
Charge entered?	false	Needed for sign-off checks.

When a nudge might be useful

STEP	WHEN THIS HAPPENS	GUIDE FIELDS TO CHECK	USUAL KIND
Intake	A case is opened for a hospital	Video Eval Location, Wake Up Protocol	Info
History	Last known well set to Unknown	Wake Up Protocol	Advisory
History	A clot-busting drug allergy is entered	Thrombolytic	Advisory
Imaging	CTA result being entered	CT Perfusion, AI (the hospital's scan-reading software)	Info
Imaging	Large vessel occlusion set to Yes	LVO Protocol (at local time)	Advisory
Recommendation	Thrombolytic given set to Yes	Thrombolytic, Thrombolytic Mix Model, Order Entry, Order Set Name	Advisory
Recommendation	"Thrombolytic order placed in the hospital's system?" shown	Order Entry, Order Set Name	Advisory
Recommendation	Seizure or EEG mentioned in the plan	EEG (at local time)	Info
Note	A note is being created	Required Note Title	Advisory
Sign-off	Complete is clicked	Charge Entry, Consult Completion Rule	Hard stop
Rounding	A rounding visit is opened	Rounding Timing, Rounding Orders	Info

You may add, remove or change these, as long as you can explain why.

The six statuses

STATUS	WHEN	WHAT THE DOCTOR SEES
Answered	The field exists and is current.	The value, with its source field.
Not in guide	This hospital's guide doesn't have the field.	"Not available in the hospital guide." Never filled from another hospital.
Time-dependent	The answer depends on the local day or time.	The answer for right now, and the rule used to decide it.
Stale	Last checked more than 180 days before the case.	The value, clearly flagged as possibly out of date.
Conflict	The guide and the storyboard disagree.	Both values with their sources. Never silently pick one.
Hard stop	A required step isn't done.	What's missing and what to do.

If a time-dependent rule mentions hours the guide never defines (like "staffed hours" with no hours given), say so. Don't assume a window.

How it should behave

- **Fast:** a nudge appears within half a second of the field change. Load the hospital's guide when the case opens, not every time a field changes.
- **Never in the way:** if guide data isn't available, the caselog keeps working and the doctor is told guidance is unavailable.
- **Read-only:** nothing the doctor does changes the guide.
- **Data, not code:** adding a hospital or a field is a data change.

CHAPTER 6

The data

This is the practice data already in your kit, not the test data. Seven made-up hospitals with deliberately different set-ups, in four US time zones. For checking whether information is out of date, the case date is **March 17, 2026**. Everything is made up. Don't use real patient, hospital or Sevaro data of any kind.

FILE	WHAT'S IN IT
hospital_guides.json	All seven guides: one record per field, with its value and the date it was last checked. Each has 21 to 25 fields, 33 different field names in all (a real guide has more than 80). Some fields are missing on purpose.
storyboard_notes.json	The second source: storyboard values that may agree or disagree with the guide.
caselog_field_map.csv	Caselog steps, fields, triggers and the guide fields they map to.
encounter_context_example.json	An example of the case details your tool receives.
validation_set.csv	Sample questions V01 to V20, with the expected status and answer.
nudge_scenarios.json	Sample cases N1 to N9, with the nudge a good tool should show (or not show).

Use the sample questions and cases to check your work, but don't build your tool around their answers. The test uses a hospital that isn't in this data.

CHAPTER 7

Going further

Not required, but worth thinking about. Pick one or two that matter to your idea and factor them in.

- **Cost:** what the mistakes cost today (doctor and quality-team time fixing them after the case, missed charges that never get billed, minutes lost looking things up mid-case), compared with what your tool would cost to run: hosting, data storage, connecting it to Synapse and hospital systems, and keeping every guide up to date.
- **Fitting into Synapse:** how this would really sit inside the caselog doctors already use, and alongside the hospital's own record system.
- **Quality:** the brief's mistakes (wrong order set, wrong note title, missed charge, wrong escalation) are caught today by a quality team after the case. How would you show they're going down?
- **Keeping guides current:** who updates a guide when a hospital changes its rules, and how the tool finds out.
- **Patient safety:** what happens if a nudge is wrong, and how the doctor can tell how much to trust it.
- **Patient privacy:** HIPAA, the US health privacy law. Case details are patient information.
- **A record:** which nudges were shown, ignored or acted on.

CHAPTER 8

Words to know

Consult

One hospital's request for a neurologist for one patient.

Teleneurologist

A neurologist treating patients remotely, by phone and video.

Synapse

Sevaro's platform for calls, video, notes and charting.

Caselog

The step-by-step form in Synapse where the doctor records the case.

Hospital guide

A hospital's own rules, one field at a time, each with the date it was last checked.

Storyboard

A short summary card for the hospital at the top of the caselog. A second source that can disagree with the guide.

Nudge

A rule shown to the doctor without them asking: hard stop, advisory or info.

Hard stop

A nudge that blocks the doctor until they fix something or acknowledge it.

Thrombolytic

A clot-busting drug. Hospitals use different ones: Alteplase (tPA) or Tenecteplase (TNK).

LVO

Large vessel occlusion: a blocked major artery in the brain, often needing a procedure to remove the clot or a transfer.

Thrombectomy (EVT)

A procedure to remove the clot. Not every hospital can do it, and some only at certain times.

Last known well

The last time the patient was known to be normal. If it's unknown (for example they woke up with symptoms), some hospitals follow a wake-up protocol.

CT, CTA, CT perfusion

Brain scans used in a stroke case. Not every hospital offers every scan at every hour.

Order set

A ready-made group of orders in the hospital's own system, found by searching its name.

Charge entry

Recording the billing charge for the visit. Some hospitals require it.

Rounding

Scheduled follow-up visits with patients already in the hospital.

EEG

A test that records the brain's electrical activity, used for seizures. Some hospitals only offer it at certain times.

ED

The hospital's emergency department.

Stale

Last checked more than 180 days before the case, so possibly out of date.

HIPAA

The US law protecting patients' health information.